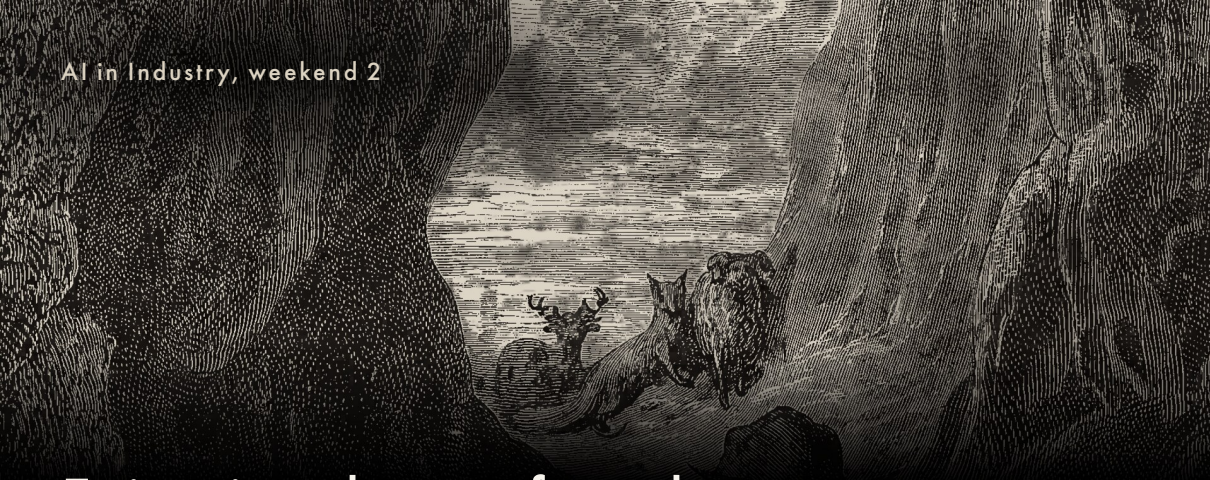
AI in Industry, weekend 2



Estimating chances from the past

A fable from ancient Greece

Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain



The lion’s cave

Ancient Greece, a fable of Aesop

An old lion can no longer hunt. He lies down in his cave and lets it be known that he is ill. The animals come to visit him, one after another.

Oudry and Chedel, *Le Lion malade et le Renard*, 1755, detail.
Universidad de Sevilla. Internet Archive, public domain



The lion's cave

Ancient Greece, a fable of Aesop

An old lion can no longer hunt. He lies down in his cave and lets it be known that he is ill. The animals come to visit him, one after another.

The fox cannot see inside the cave. How can she tell what happens to the visitors?

Oudry and Chedel, *Le Lion malade et le Renard*, 1755, detail.
Universidad de Sevilla. Internet Archive, public domain



καὶ στᾶσα ἄποθεν τοῦ σπηλαίου
ἐπυνθάνετο αὐτοῦ πῶς ἔχοι.

She stood at a distance from the cave and asked him how he was.

Aesop, *The old lion and the fox*, Chambry 196.
English: lecturer

Wenceslaus Hollar, *The sick lion*, in Ogilby's Aesop, 1668, detail.
University of Toronto. Wikimedia Commons, public domain



Wenceslaus Hollar, *The sick lion*, in Ogilby's *Aesop*, 1668, detail. University of Toronto. Wikimedia Commons, public domain

The lion

too old to hunt

"Badly," he said, and he asked her why she did not come in.

Aesop, *The old lion and the fox*, Chambry 196. English: lecturer

The fox

at a distance from the cave

"But I would have come in, if I did not see the tracks of many going in, and of none coming out."



Aesop, *The old lion and the fox*, Chambry 196.
English: lecturer

Oudry and Chedel, *Le Lion malade et le Renard*, 1755, detail.
Universidad de Sevilla. Internet Archive, public domain



"So sensible people foresee dangers from the evidence and escape them."

Aesop, *The old lion and the fox*, Chambry 196.
English: lecturer

Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain



"So sensible people foresee dangers from the evidence and escape them."

The fox cannot see inside the cave.
She counts what happened before: many went in, none came out.
From the past, she estimates her own chance of coming out.

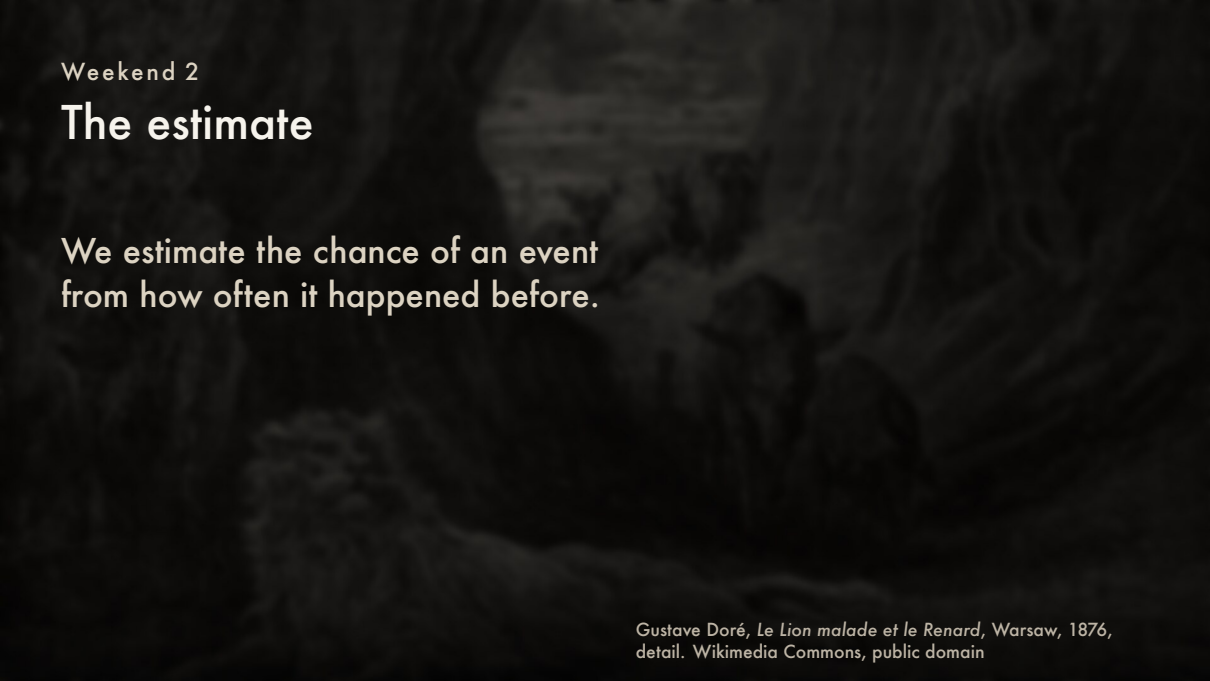
Aesop, *The old lion and the fox*, Chambry 196.
English: lecturer

Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain

Weekend 2

The estimate

We estimate the chance of an event from how often it happened before.



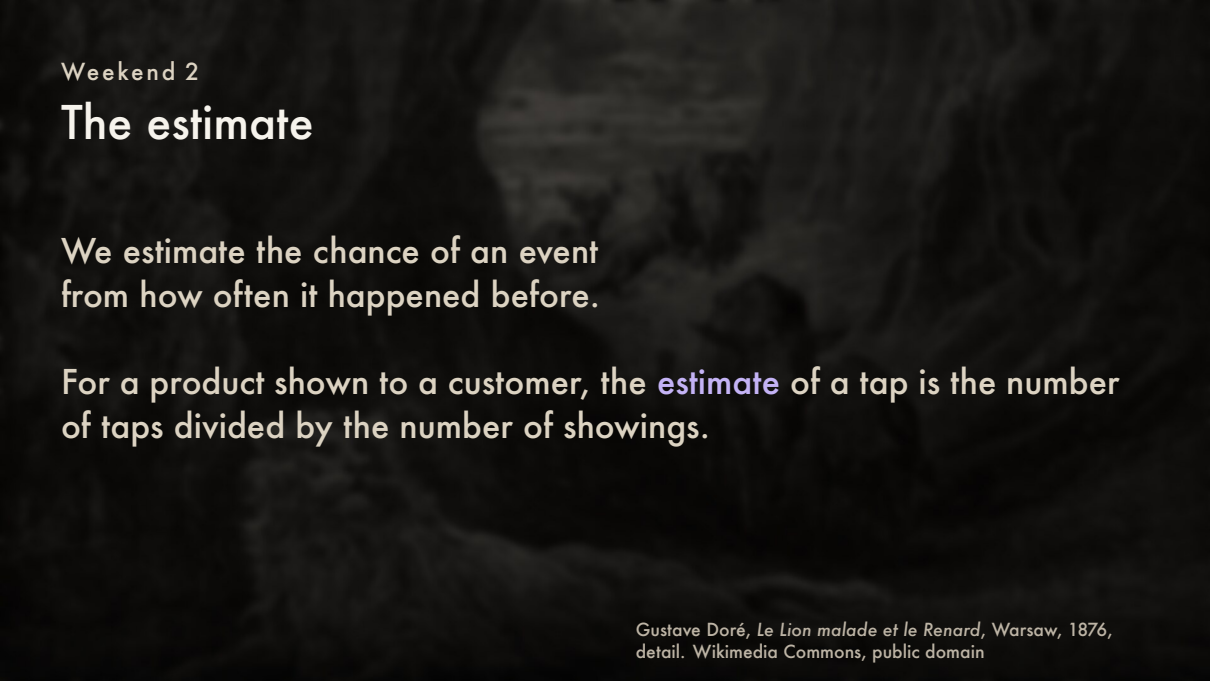
Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain

Weekend 2

The estimate

We estimate the chance of an event from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.



Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain

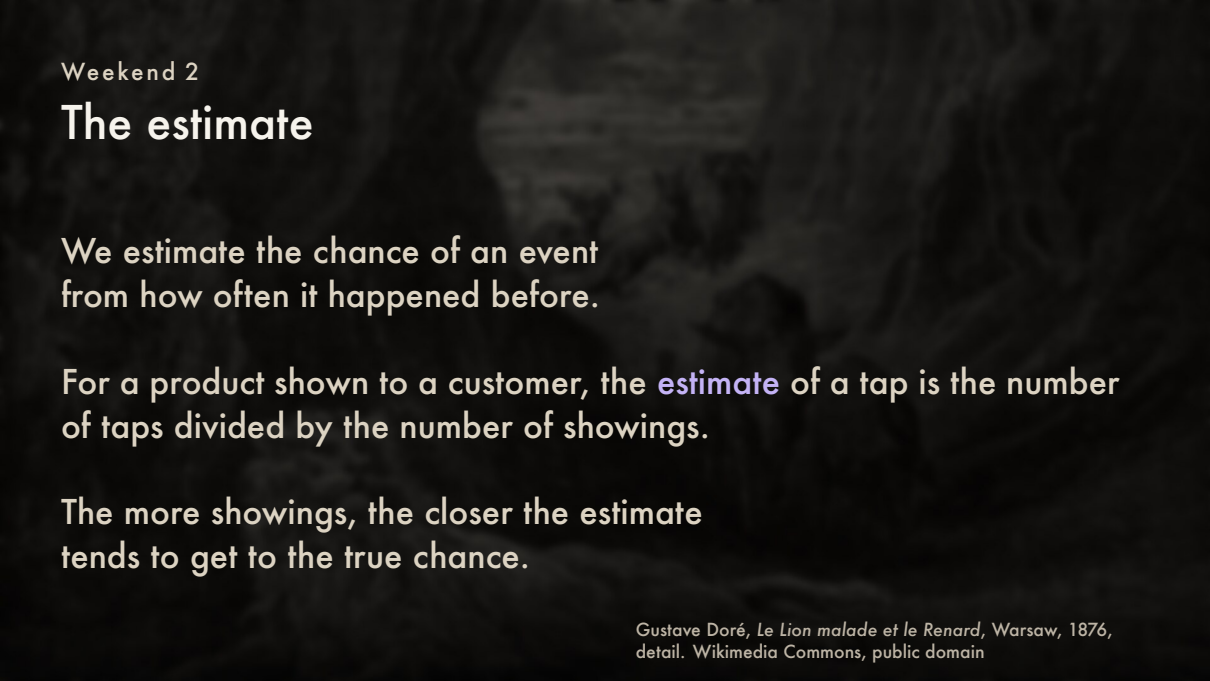
Weekend 2

The estimate

We estimate the chance of an event
from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.

The more showings, the closer the estimate
tends to get to the true chance.



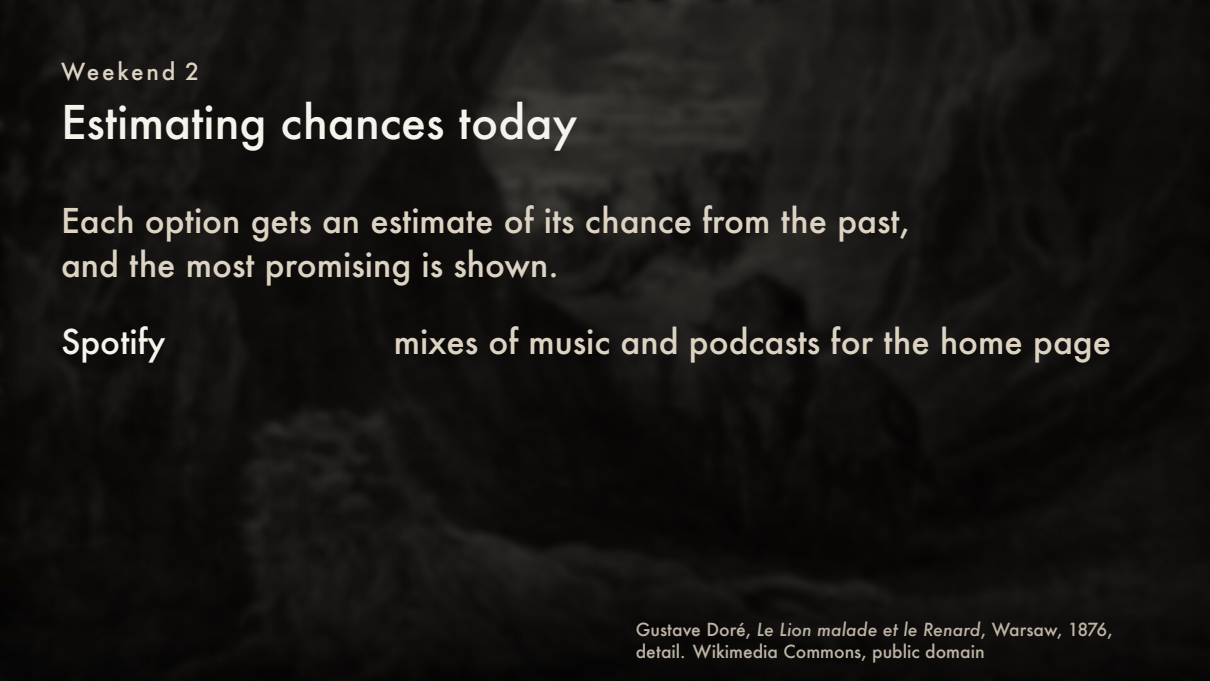
Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain

Weekend 2

Estimating chances today

Each option gets an estimate of its chance from the past,
and the most promising is shown.

Spotify mixes of music and podcasts for the home page



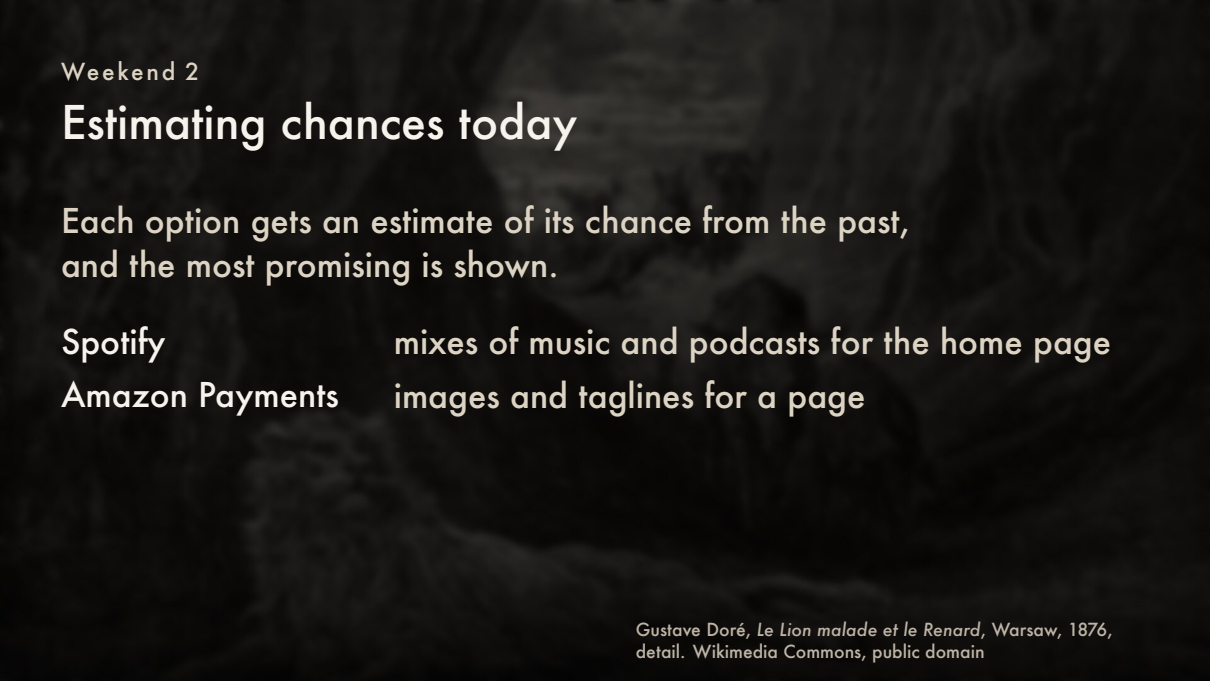
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Weekend 2

Estimating chances today

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Spotify	mixes of music and podcasts for the home page
Amazon Payments	images and taglines for a page



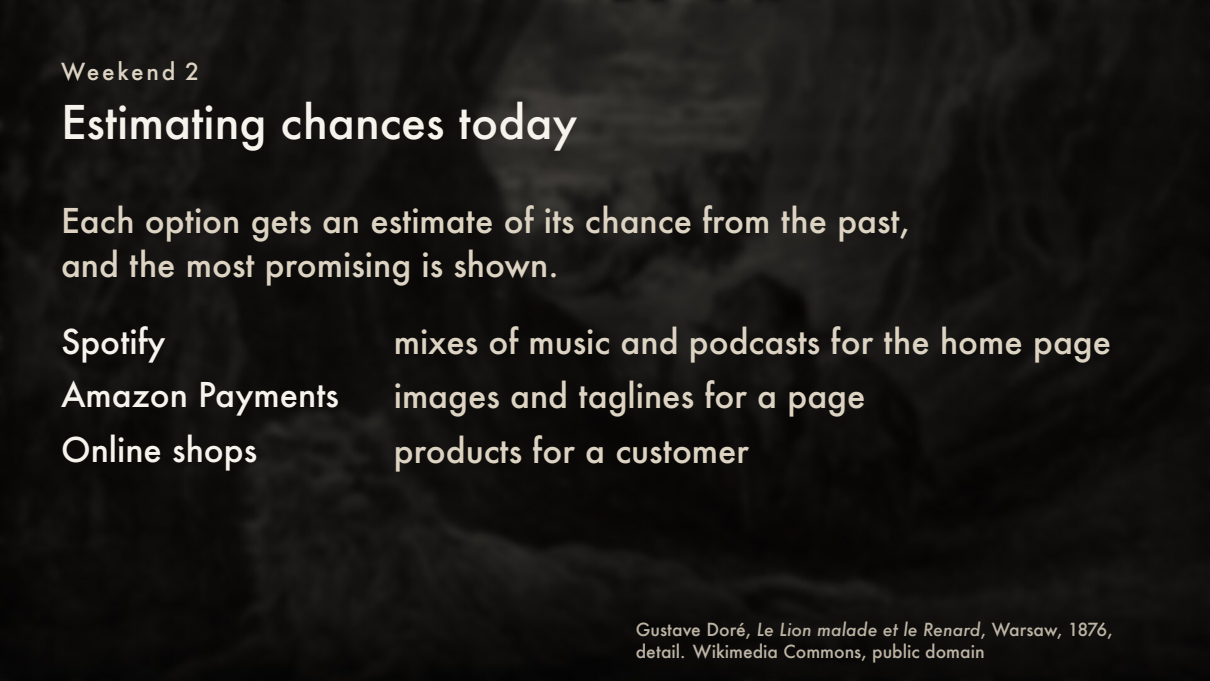
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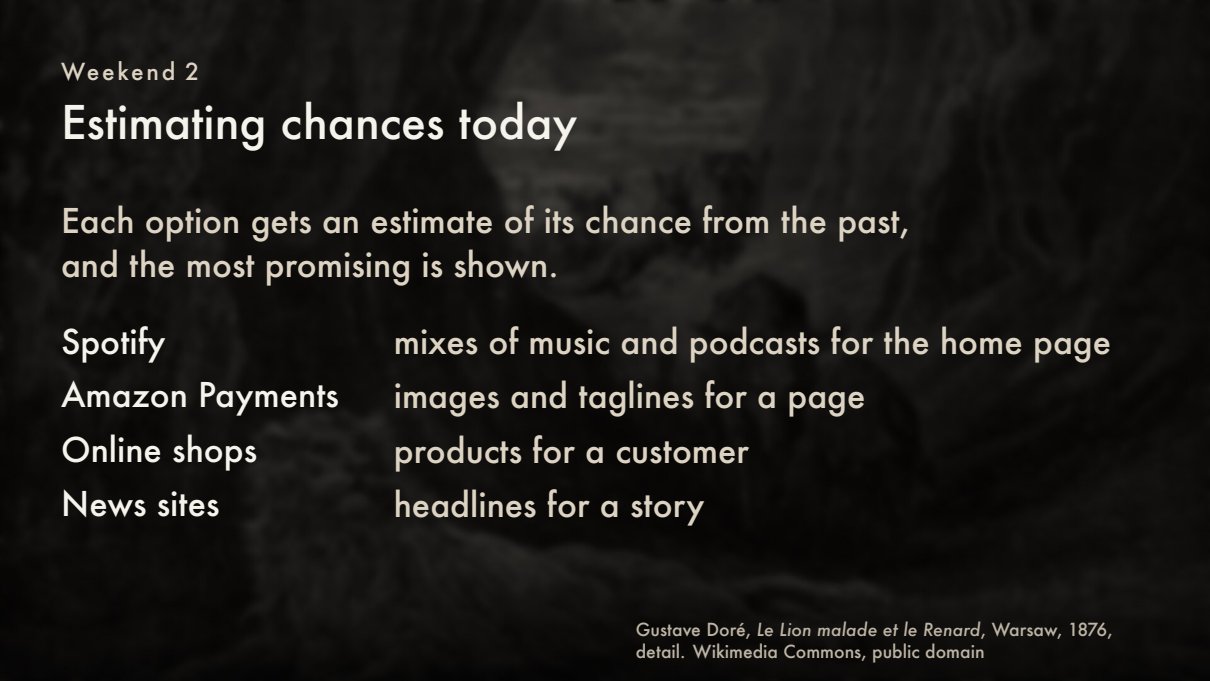
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Weekend 2

Estimating chances today

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Amazon Payments	images and taglines for a page
Online shops	products for a customer
News sites	headlines for a story



Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876, detail. Wikimedia Commons, public domain

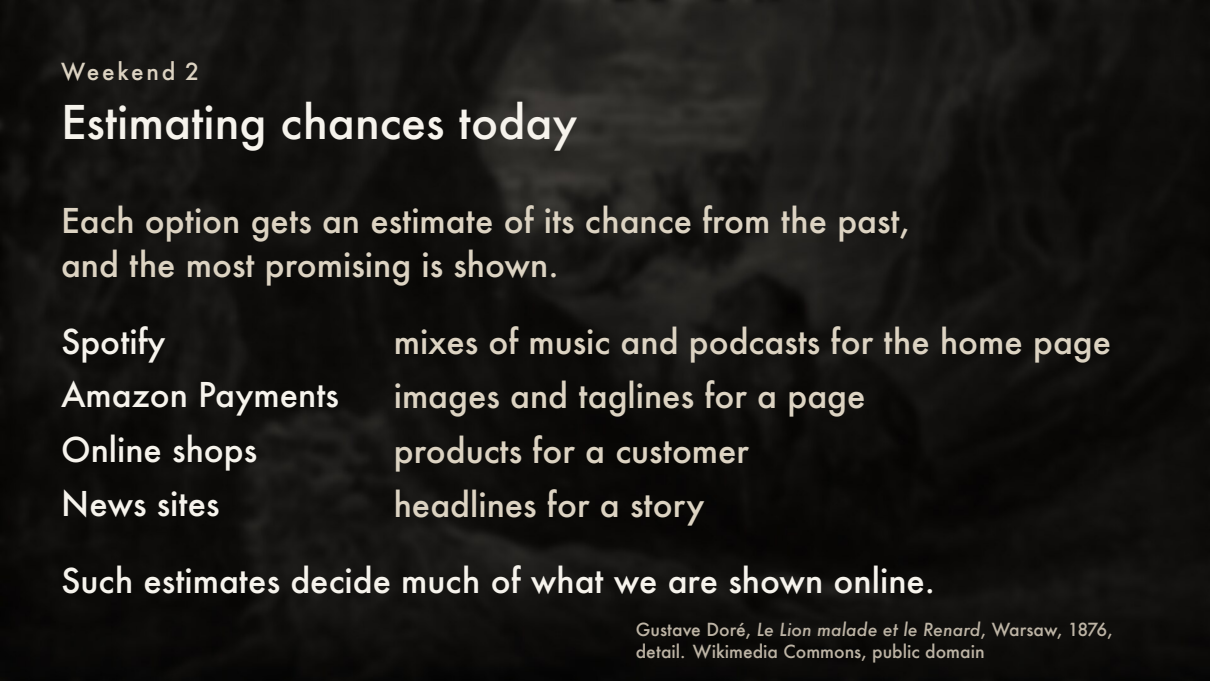
Weekend 2

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Amazon Payments	images and taglines for a page
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Such estimates decide much of what we are shown online.



Gustave Doré, *Le Lion malade et le Renard*, Warsaw, 1876,
detail. Wikimedia Commons, public domain